

Tour into the Picture

Computer Vision 2024 – Group 29 | Luca Martin, Hassan R. Khan, Fatma Tej, Yi Yue, Philipp Wang, Ali Rabeh

Motivation

In the course, Computer Vision, we learnt methods of 3D-reconstruction using two camera perspectives. The Tour Into The Picture method offers a basic geometric approach to recreate an immersive scene using a single image.

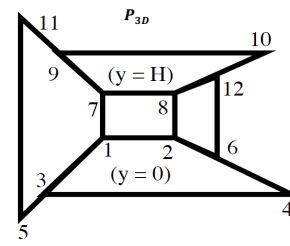
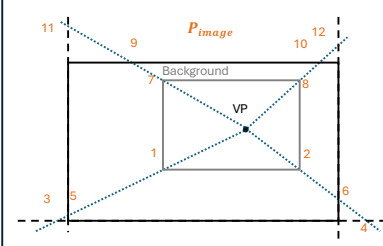
1 Foreground selection and patch-based inpainting

- Users can mark objects present in the scene (so-called *target regions*).
- The algorithm identifies target regions using a binary mask.
- An exemplar-based search computes sum of squared differences to evaluate similarity between target and source region (*rest of the image*).
- Due to parallel computing support, provides advantage of diffusion-based techniques in terms of performance.



2 Calculation 12 Points in image- and world-coordinates

1. User selects background and vanishing point
2. Calculate 12 points in image Coordinates
3. Calculate 12 points in world coordinates



7 Improvement opportunities & future work

Problems with our method:

1. Handle Images with two vanish points
2. Handle images with no clear background
3. Foreground objects can only be placed on the ground for now
4. We can effectively construct a 3D space, we can only separate the foreground.
5. it cannot obtain depth information (3D information) in the background.
6. Essentially, it only displays projections from different perspectives.

Future Work:

We can use the following methods

1. Extension to Panoramic views
2. Structure from Motion (SfM)
3. Deep learning methods: Depth Estimation Neural Radiance Fields, NeRF

6 Results of 3D-Scene



3 Calculation of homography-matrices

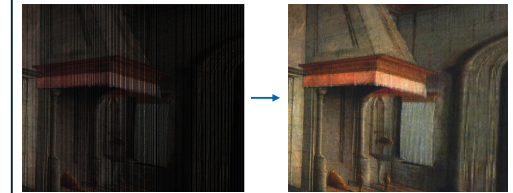
- The homography-matrix transforms the points from one plane to another:

$$\begin{bmatrix} P_{3D} \\ 1 \end{bmatrix} = H \begin{bmatrix} P_{image} \\ 1 \end{bmatrix}$$

- P_{image}/P_{3D} consists of 4 Points respectively (corners of each face)
- Five different homography matrices are calculated - one for every face
- Since all points of P_{3D} and P_{image} are known, H can be calculated by solving the linear equation (8 unknowns)

4 Interpolation

- The homographic transformation via H leads to gaps in the transformed image
- The gaps are filled by using *regionfill*. It uses the surrounding pixels to find the color for the gaps.



5 Modeling foreground objects

- Challenge: Extracted foreground object has unwanted marginal area => Create new mask to improve foreground object
- Positioning: Consider the object as a rectangle to construct similar triangles



References

- https://docs.opencv.org/4.x/d9/dab/tutorial_homography.html - last visited 09.07.2024
- Youichi Horry, Ken-Ichi Anjiyo, and Kiyoshi Arai. Tour into the picture: using a spidery mesh interface to make animation from a single image
- Yu, A., Ye, V., Tancik, M., & Kanazawa, A. (2021). pixelnerf: Neural radiance fields from one or few images. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 4578-4587).
- Hesabi, Somayeh. (2012). A Modified Patch Propagation-based Image Inpainting Using Patch Sparsity. 10.13140/2.1.3915.8722.